Caspar Controller

Review and recommendations.

1. Summary

Caspar Controller lets one Windows PC run several graphics outputs at once and send each one over the network (NDI) to vision mixers such as the Tricaster. Operators control it from a web page, and hardware panels such as Companion can trigger clips.

The app works and is useful, but **it is not ready to be rolled out across the company** and has three problems that need fixing first:

- **Security:** anyone on the same network can take control of the PC it runs on.
- **Deployment:** it can't be installed properly through Intune, and it updates itself from a personal GitHub account.
- **Maintainability:** the code is hard to change safely and has no automated tests.

Recommendation: keep the product as it is for users (web interface, clip triggering, control of other machines) and rebuild the internals in five phases. The first phase fixes the security issues and can start straight away.

Immediate action: until Phase 0 is complete, don't install the current version on any more machines, and only use it on isolated production networks.

2. Recommendations

2.1 Recommended approach

Each playout PC runs a small background program (an "agent") that manages that PC's graphics outputs. The agent:

- Provides the **web interface**, which works from any browser on the network.
- Provides a **secure API**, protected by an API key, for Companion and hardware panels.
- Can **see and control the other playout PCs** from one screen (Remote Controllers).
- Shows a **tray icon** on the PC so operators can see it's running.

Two other options were considered and rejected:

- **A desktop-only app:** it would lose clip triggering and remote control, which the galleries need.
- **A separate desktop app plus a web version:** two interfaces to build and maintain, which is the duplication we're removing.

If a central dashboard is ever wanted, it can be added later on top of the same API without rework.

2.2 What users will get

Improvement	Benefit
One-click install from Company Portal	No manual downloads, zip files or file paths. CasparCG and the NDI Runtime install automatically.
Setup wizard on first launch	New users get from install to working outputs in a few steps.
Console panel in the web interface	Engineers can see each output's diagnostic log in one place instead of hidden windows.

Improvement	Benefit
Automatic restart (on by default)	If an output crashes or freezes, it restarts and reloads its graphics without anyone touching it. It can be switched off per output.
Secure clip triggering	Companion and panels keep working, now with an API key so only approved devices can trigger clips.
Instant status updates	The dashboard updates live instead of every few seconds.
Smooth upgrade for existing users	Settings on machines already running the app are carried over automatically.

2.3 Name

Rename the app to **Caspar Controller**, dropping the original author's name and the mixed Casper/Caspar spelling.

The new name is used everywhere users see it: the installer, Company Portal, the tray icon and the web interface. The code moves to the company GitHub organisation under the same name, and the bundled CasparCG becomes **CasparCG Server (ITN build)**. The original author is credited in the app's About box.

2.4 Delivery plan

The work is split into five phases.

Phase	Outcome	Main deliverables	Size
0. Urgent fixes	Safe to run on a network	API key and login on everything, close the security holes, switch off self-updating	Small
1. Restructure	Reliable and testable	Automated tests of current behaviour, code reorganised, one component in charge of the outputs	Large
2. Web interface	Better operator experience	Console panel, automatic restart, live updates, improved trigger page, safer remote control	Medium
3. Deployment	Company Portal install	Signed installer, CasparCG as its own Intune package, migration for existing users, setup wizard	Medium
4. Polish	Supportable long-term	Log files, health monitoring, optional limited "trigger-only" keys, documentation	Small

3. Review findings

The review covered the app's code, installer and build process. Findings are grouped by theme and explained in non-technical terms. The full technical list, with file and line references, is in docs/REVIEW.md.

Theme	High	Medium	Low
Security	4	5	0

Theme	High	Medium	Low
Reliability	1	5	3
Deployment	2	2	2

A further ten, more detailed issues were found in recently changed features (mainly the clip and status display) and are included in the technical document.

A separate pass looked at unnecessary complexity and found about **2,000 lines of code that can be removed**, around a third of the project.

3.1 Security

Finding	Severity	What it means
The app's controls need no password	High	Anyone on the network can start or stop outputs, change settings or send commands.
A crafted output name can run programs on the PC	High	Combined with the point above, anyone on the network could take over the playout PC.
Files on the PC can be read through the web address	High	Any file on the machine, including the app's settings, can be downloaded.
The web password is exposed	High	The password can be read back by anyone, so it gives no real protection.
Other weaknesses	Medium	The app can be used to reach other internal systems, imported settings aren't checked, and some inputs aren't cleaned.

3.2 Reliability

Finding	Severity	What it means
Starting the app can close unrelated programs	High	It shuts down any CasparCG on the machine, and anything connected to its web port.
Status can show an output as off when it is on air	Medium	An operator pressing Start could restart a live output.
Saving settings during a show can affect the wrong output	Medium	Reordering outputs while live can make Stop act on a different output.
Settings can be lost or corrupted	Medium	Saving isn't done safely, so a crash or busy moment can damage the settings file.
On-air (tally) indicators can be wrong	Medium	Similar output names, or the same name on another PC, can light the wrong card.
Diagnostic consoles are hidden and crashes aren't noticed	Medium	Engineers can't see why an output failed, and nothing restarts it.
No log files and some slow responses	Low	Problems after an incident are hard to investigate.

3.3 Deployment

Finding	Severity	What it means
Installer doesn't suit Intune	High	Installed through Intune, the app would be hidden from users and unable to save settings.
App updates itself from a personal GitHub account	High	Unchecked software could be installed silently, outside IT's control.
Installer downloads 500 MB during install	Medium	Installs fail without internet access, and the CasparCG version isn't controlled.
Software isn't digitally signed	Medium	Windows and antivirus may warn about or block it.
Licensing and branding not checked	Low	Bundled fonts and CasparCG need licence checks; the original author's name and gallery names are hard-coded.

3.4 Maintainability

- The same start and stop logic is written twice, once for the desktop window and once for the web interface. The two copies have already drifted apart.
- Two very large files hold most of the app, including the whole web interface written inside the code.
- There are no automated tests, so every change has to be checked by hand.
- Old scripts, test folders and unused features are still in the project.
- The setup instructions are out of date and involve around 15 manual steps.